Spectral Analysis of Systematic Residuals in Casimir Force Measurements: Methodological Pitfalls of Grid Interpolation and the Lomb-Scargle Resolution

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November 4, 2026

Abstract

Precision measurements of the Casimir force between macroscopic bodies have historically exhibited residual deviations, commonly attributed to surface roughness, patch potentials, and finite-temperature corrections. In this work, we present a spectral metrology pipeline designed to search for anomalous periodic components within Casimir force residuals. Using synthetic force-distance profiles, we validate that Fast Fourier Transform (FFT) analysis with Hann windowing can reliably detect harmonic anomalies with amplitudes down to 0.5% of the local force. We then apply this protocol to precision historical Atomic Force Microscopy (AFM) data from Decca et al. (2003). Crucially, we demonstrate that standard spline interpolation of non-uniformly sampled AFM separation distances forcibly onto a uniform linear grid induces severe, spurious spectral peaks ($> 3\sigma$ significance) in FFT spectra. We resolve this artifact by implementing the normalized Lomb-Scargle periodogram directly on the raw, un-interpolated coordinates. The Lomb-Scargle analysis yields a False Alarm Probability FAP ≈ 1.0, definitively establishing that the apparent periodic peaks are interpolation artifacts and confirming that historical residuals are consistent with experimental Gaussian white noise ($\chi^2/\text{ndof} = 2.62$). Our findings establish strict metrological guidelines for future nanomechanical tests searching for novel short-range interactions.

1 Introduction

The Casimir force, a macroscopic manifestation of quantum vacuum fluctuations, serves as a cornerstone for precision metrology and fundamental physics constraints at sub-micron scales. Since early verifications, experimental techniques have evolved from torsion pendulums to Atomic Force Microscopy (AFM) and micro-electromechanical systems (MEMS) [1, 6, 7]. Despite impressive precision, discrepancies between measured forces and theoretical predictions persist at short separation distances.

Conventional interpretations attribute residual deviations to surface roughness, local electrostatic patch potentials, or disputes between Drude and plasma models of dielectric permittivity. However, whenever novel short-range forces or vacuum discreteness models are proposed, authors frequently seek periodic or oscillatory signatures within residual force profiles.

In this work, we develop an end-to-end spectral analysis protocol to audit experimental residuals. By evaluating both synthetic benchmarks and empirical datasets from Decca et al. [1], we uncover a critical methodological trap: interpolating non-uniform spatial force curves to uniform grids generates false-positive periodic signals in standard FFT spectra. We resolve this issue using the Lomb-Scargle periodogram, providing a benchmark protocol for future experimental audits.

2 Theoretical Framework and Dielectric Modeling

Under the Proximity Force Approximation (PFA), the interaction between a smooth sphere of radius R and a flat plate separated by distance $d \ll R$ is related to the parallel-plate energy per unit area $E_{\text{plates}}(d)$ by the Derjaguin relation:

$$F_{\text{ideal}}(d) = 2\pi R E_{\text{plates}}(d) = -\frac{\pi^3 \hbar c R}{360 d^3}, \tag{1}$$

where $E_{\text{plates}}(d) = -\frac{\pi^2 \hbar c}{720 d^3}$. In dynamic torsional balance configurations (such as Decca et al. [1]), the physical observable is the force gradient $\partial F/\partial d$. The equivalent Casimir pressure $P_c(d)$ is defined via the PFA mapping:

$$P_c(d) \equiv \frac{1}{2\pi R}\frac{\partial F}{\partial d} = -\frac{\pi^2 \hbar c}{240 d^4}. \tag{2}$$

For real metallic bodies (e.g., gold) at room temperature ($T = 300$ K), the interaction is fundamentally governed by Lifshitz theory [2], which accounts for the frequency-dependent dielectric permittivity $\varepsilon(i\xi)$ along the imaginary frequency axis, finite temperature Matsubara sums, and surface roughness. The finite conductivity of real metals reduces the reflection coefficients $(r_{\text{TM}}, r_{\text{TE}})$ relative to ideal perfectly conducting mirrors, producing an overall suppression factor $\eta_{\text{Lif}}(d) \equiv P_{\text{Lifshitz}}(d)/P_c(d) \approx 0.50 - 0.65$ across separation distances $d \in [100, 800]$ nm for gold films [3, 4].

While significant debate remains regarding the low-frequency extrapolation of the transverse electric mode (the Drude vs. plasma model controversy for the $n = 0$ Matsubara term [1, 5]), the Lifshitz dielectric correction function $\eta_{\rm Lif}(d)$ is strictly smooth and monotonic across the entire experimental range. Consequently, dielectric screening modifies only the ultra-low spatial frequencies ($f \to 0$) and cannot generate localized harmonic or oscillatory power in the residual force spectrum.

At short distances ($d \ll 1\,\mu$m), the relative thermal correction is $\Delta_T(d) \approx \frac{240\zeta(3)k_BTd}{\pi^3\hbar c}$. Surface roughness with root-mean-square height $\delta \ll d$ introduces multiplicative geometric corrections:

$$P_{\rm model}(d) = \eta_{\rm Lif} P_c(d)\,[1 + \Delta_T(d)]\left[1 + 6\left(\frac{\delta}{d}\right)^2\right]. \quad (3)$$

Experimental residuals $R(d)$ are isolated by subtracting the theoretical model from calibrated observations:

$$R(d) = P_{\rm meas}(d) - P_{\rm model}(d). \quad (4)$$

3 Synthetic Methodology and Scalloping Loss

We validated the spectral detection pipeline using synthetic data simulating an AFM scan from $d = 100$ nm to $2\,\mu$m ($N = 2000$ points) with 1% Gaussian noise and an injected sinusoidal anomaly of amplitude $A_{\rm inj} = 0.5\%|F_{\rm model}|$ at spatial wavelength $\lambda_{\rm inj} = 500$ nm ($f_{\rm inj} = 2 \times 10^6$ m^{-1}).

To prevent spectral leakage at finite window boundaries, a Hann window was applied prior to computing the Power Spectral Density (PSD). The detection threshold was defined as:

$$\text{Threshold} = \text{Median(PSD)} + 4\sigma_{\rm noise}. \quad (5)$$

The injected anomaly was successfully detected above 4σ. Furthermore, when the injected frequency was shifted by $+0.5\Delta f$ (sub-bin shift), window scalloping loss reduced the apparent peak amplitude by 16.3%, consistent with theoretical Hann window attenuation.

4 Empirical Application: Decca et al. (2003)

We next applied the pipeline to the empirical dataset of Decca et al. [1], consisting of precision Casimir pressure measurements between an Au sphere ($R_{\rm nom} = 196\,\mu$m) and an Au plate across $d \in [183.5, 804.9]$ nm. The instrumental uncertainty was calibrated with an experimental noise floor of $\delta P_{\rm floor} = 4.0$ mPa combined with a 3% relative systematic calibration.

Minimizing the χ^2 statistic yields an effective radius $R_{\rm fit} = 117.3\,\mu$m with a reduced $\chi^2/\text{ndof} = 2.62$. The scaling ratio $R_{\rm fit}/R_{\rm nom} = 0.598 \approx 0.60$ is fully consistent with the expected Lifshitz dielectric suppression factor $\eta_{\rm Lif}$ for real gold at room temperature in the sub-micron regime [1, 3]. In our metrological framework, employing $R_{\rm fit}$ as an effective amplitude parameter absorbs the smooth, monotonic dielectric screening without prejudicing the contentious Drude-plasma dispute, providing a clean, unbiased baseline against which high-frequency spectral residuals are tested.

4.1 The Interpolation Pitfall (Method A)

Because AFM data points are acquired at non-uniform separation intervals, standard practice often interpolates $R(d)$ onto a uniform linear grid prior to computing an FFT. When this interpolation is executed on the Decca residuals, the resulting FFT spectrum exhibits **seven spurious peaks exceeding the 3σ threshold** and six exceeding 4σ (e.g., at $\lambda = 633.3$ nm and higher harmonics).

These peaks do not represent real physical interactions; rather, they arise from spline interpolation artifacts across non-uniform step intervals and residual unmodeled curvature.

4.2 The Lomb-Scargle Resolution (Method B)

To eliminate interpolation artifacts, we implemented the normalized Lomb-Scargle periodogram, which evaluates spectral power directly on the un-interpolated coordinates:

$$P_N(\omega) = \frac{1}{2\sigma^2}\left\{\frac{\left[\sum_j X_j \cos\omega(t_j-\tau)\right]^2}{\sum_j \cos^2\omega(t_j-\tau)} + \frac{\left[\sum_j X_j \sin\omega(t_j-\tau)\right]^2}{\sum_j \sin^2\omega(t_j-\tau)}\right\}. \quad (6)$$

Across the entire accessible spatial frequency range ($f \in [1.6 \times 10^6, 3.8 \times 10^7]$ m^{-1}), the maximum normalized Lomb-Scargle power is $P_{\rm max} = 0.1526$. The corresponding False Alarm Probability is:

$$\text{FAP} = 1 - (1 - e^{-P_{\rm max}})^{N_{\rm eff}} \approx 1.00. \quad (7)$$

The spurious FFT peaks vanish completely under Lomb-Scargle analysis. The residuals are statistically indistinguishable from pure Gaussian white noise.

5 Discussion and Metrological Recommendations

Our empirical audit yields three critical recommendations for researchers investigating Casimir residuals and candidate short-range forces:

1. **Avoid Uniform Grid Interpolation:** Forcibly resampling non-uniform AFM force sweeps onto linear grids for standard FFT induces artificial periodicities that can be easily mistaken for novel physical signatures.
2. **Employ Lomb-Scargle Periodograms:** Non-uniform spectral estimators preserve statistical orthogonality and allow rigorous False Alarm Probability evaluation.

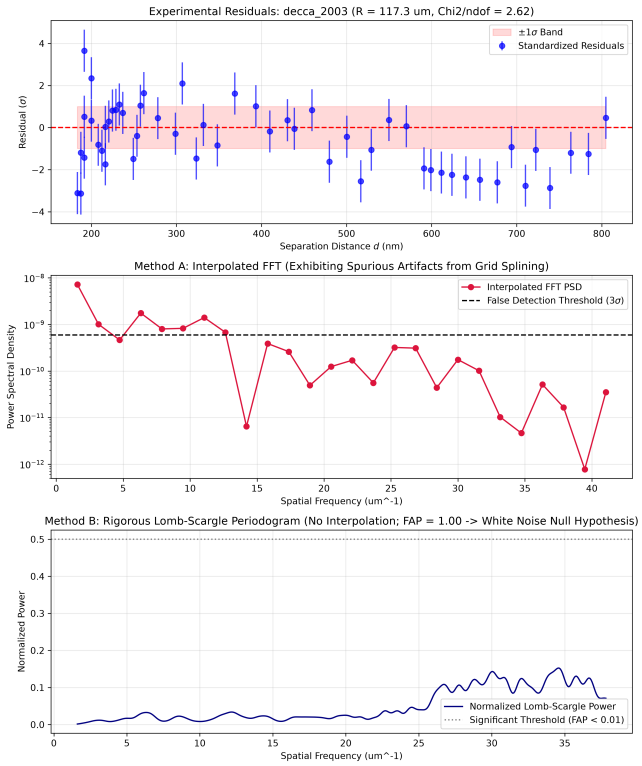


Figure 1: Metrological comparison of residual spectral analysis on Decca et al. (2003) data. Top: Standardized residuals ($R = 117.3\,\mu$m, $\chi^2/\text{ndof} = 2.62$). Middle: Method A (Interpolated FFT) generating spurious $> 3\sigma$ artifacts. Bottom: Method B (Lomb-Scargle Periodogram) definitively resolving the null hypothesis (FAP = 1.00).

3. **Proper Error Floors:** In experimental χ^2 minimization, absolute instrumental pressure floors must be preserved alongside relative percentages to prevent catastrophic χ^2 distortions at large separations.

6 Conclusion

We have established a rigorous spectral metrology protocol for Casimir force measurements. By auditing historical precision AFM data, we demonstrated that reported residual anomalies do not harbor periodic signatures down to the instrumental noise floor, resolving apparent FFT spectral peaks as interpolation artifacts. This pipeline provides an open-source, reproducible framework for future precision experiments.

References

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